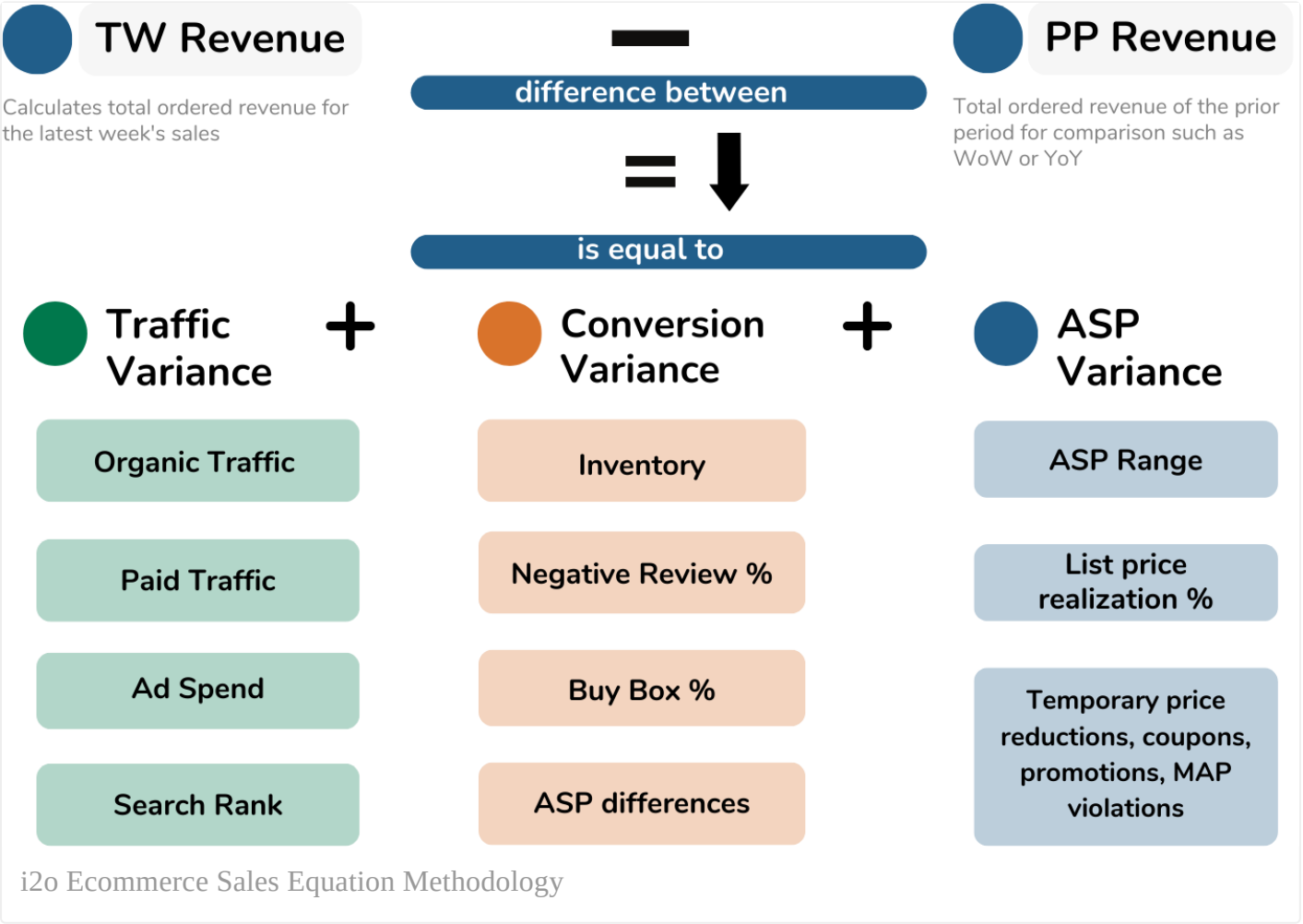
The E-commerce Sales Equation Navigator

Introduction

The **E-commerce Sales Equation (ESE) Navigator** is accessible as part of the i2o Growth Accelerator subscription. The **i2o Sales Equation** is calculated as:

Traffic x Conversion x ASP = Sales

The ESE Navigator is a sales analytics tool that compares Amazon performance across two given time periods. In this comparison, products are grouped together by sales performance on a scale of poor to excellent. Brands can track these variances to quickly identify positive and negative factors affecting Amazon sales from a broad range of metrics.

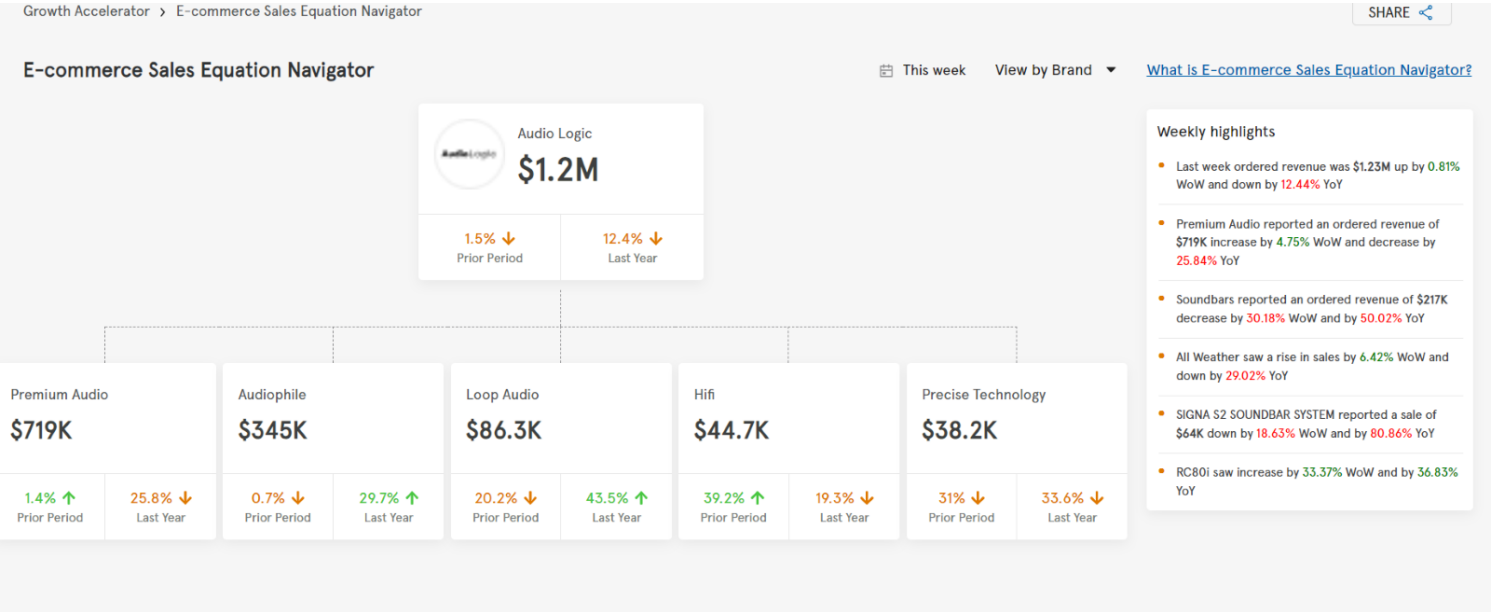


i2o Ecommerce Sales Equation Methodology

Metrics indicate factors which affect traffic, conversion, and average selling price. By comparing the current period to a prior one (such as WoW or YoY), ecommerce brands can evaluate the effectiveness of any changes made to their product offerings, pricing, or marketing strategies over time. For example, users can monitor how losing the Buy Box leads to a negative impact in sales and conversion with a quantifiable attribution.

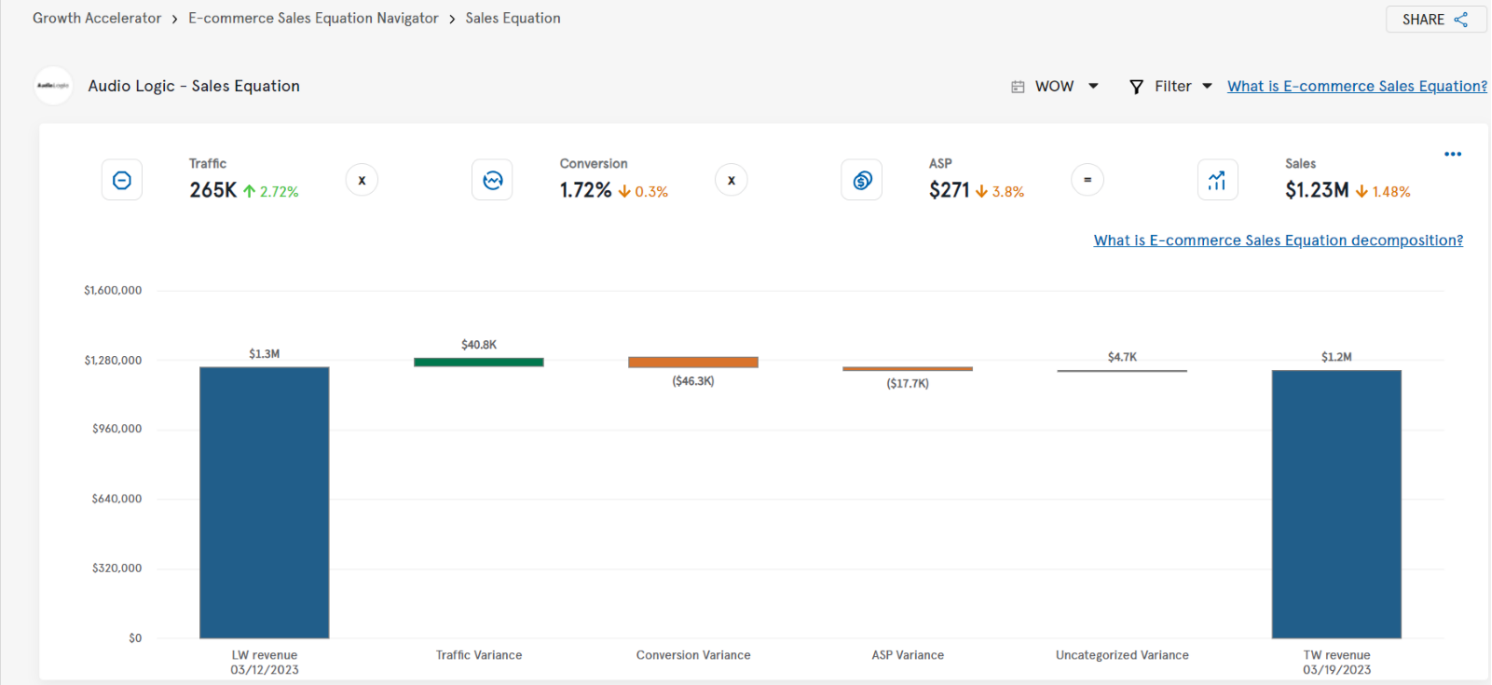
Home Page

On the home page of the ESE Navigator, the latest week's Amazon sales are shown in a top-down hierarchy from overall brand into either child brand or category. You can see the Ordered Revenue Sales ($) value broken down by brand and category. You can also see the change in sales (%) from either the prior period or the same week last year.



Sales Equation Page

You can view the change across periods depicted as a waterfall visualization. A positive sales value is displayed above the bar whereas a negative value is directed below the bar. This helps you identify whether revenue was gained or lost due to factors from traffic, conversion, or average selling price from factors like lower paid traffic, higher negative reviews, or temporary price reductions, respectively.
Using the filters on this page, the latest week of Amazon sales can be compared Week over Week (WoW), Year over Year (YoY), or against the T4 (Trailing 4-weeks) Average. The Traffic Variance, the Conversion Variance, and the ASP Variance reports open only from this waterfall chart.



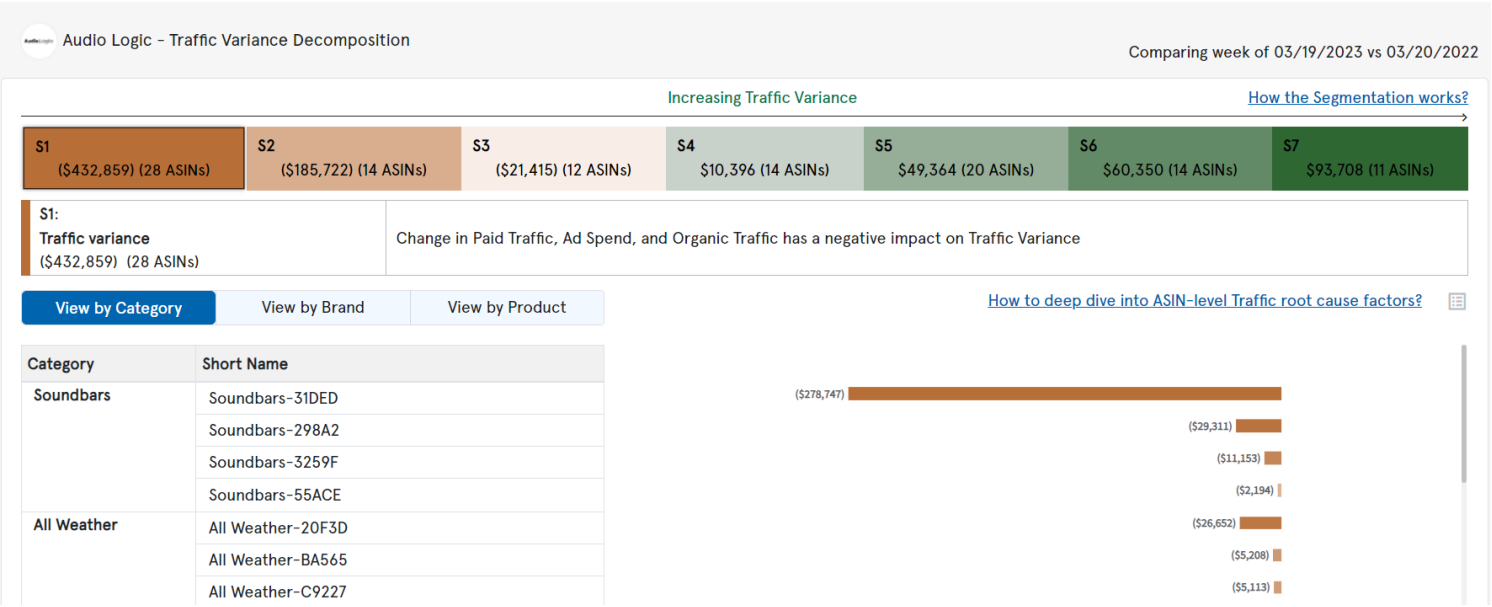
Click on the bars labeled **Traffic Variance, Conversion Variance,** or **ASP Variance** in the waterfall chart to drill into a decomposition of sales by performance ranging from poor to excellent.

Traffic Variance Page

On this page, products are placed in mutually exclusive Segments using a proprietary i2o Machine Learning Algorithm. Products within each Segment share a similar traffic factor which has affected their sales variance in the selected time range (WoW, YoY, T4).

The metrics for the Traffic Variance are **Paid Traffic, Organic Traffic,** and **Ad Spend**.

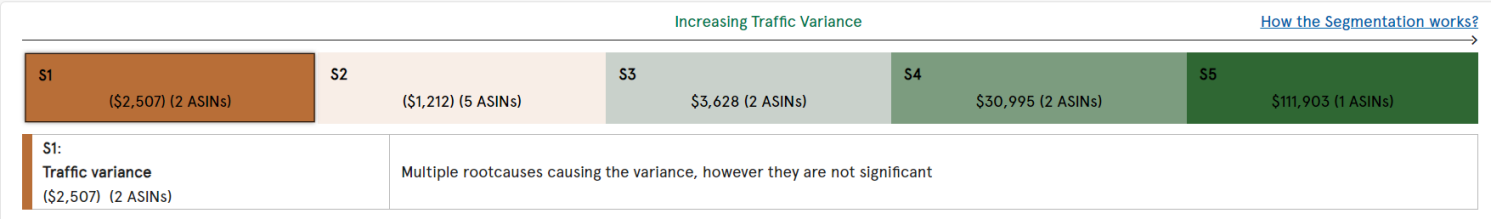
The sum of variances across all segments is equal to the Traffic Variance computed in the previous screen.



The following is an example of product analysis capabilities in the Traffic Variance. A product with a Traffic Variance of ($1,000) sees a sales decline due to a decrease in Ad Spend and Paid Traffic. This sales decline was not impacted by Organic Traffic. A different product with a Traffic Variance of ($3,000) sees a significant decline from a large drop in Organic Traffic while the Paid Traffic and Ad Spend had no impact. As a result, users can focus on Traffic metrics which are truly impacting sales.

Traffic Segments

Segments are labeled under the names S1, S2, S3, etc. in order of the most negative variance to the most positive variance.



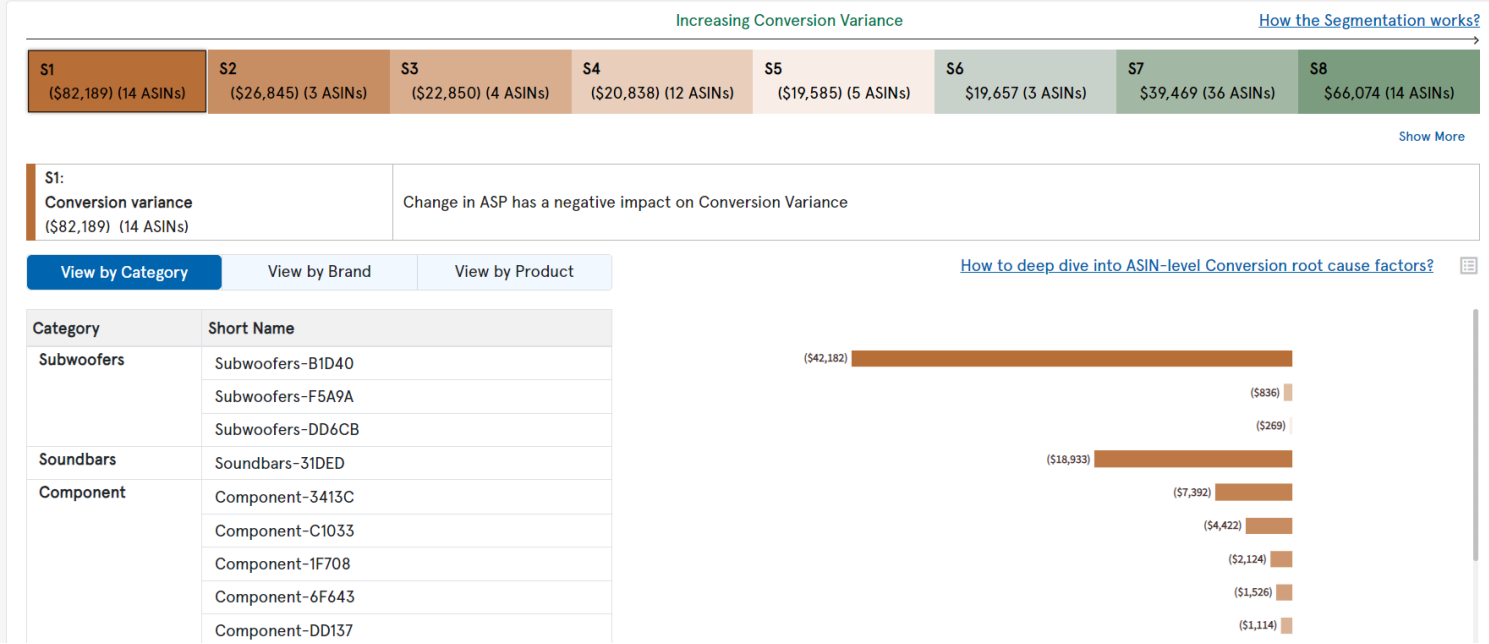
Click on any Segment of the Traffic Decomposition to see product groups from poor to excellent sales performance.

Conversion Variance Page

On this page, products are placed in mutually exclusive Segments using a proprietary i2o Machine Learning Algorithm. Products in each Segment share a similar conversion factor which has affected their sales variance in the selected time range (WoW, YoY, T4). The sum of variances across all segments is equal to the Conversion Variance computed on the previous screen.

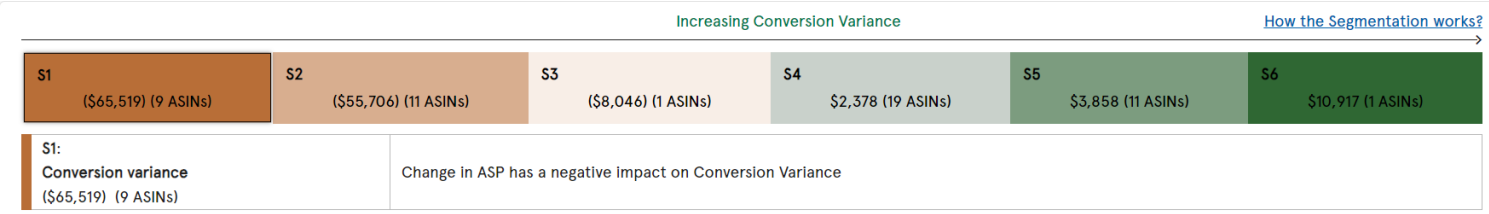
The metrics for the Conversion Variance are **Inventory, Buy Box pressure, Change in ASP,** and **Negative Review %.**

The following is an example of product analysis capabilities in the Conversion Variance. A product with a Conversion Variance of ($7,000) sees a large drop in Observed Won Buy Box percentage and a sharp drop in average selling price. A different product with a Conversion Variance of ($3,750) sees a decline after a large jump in Customer Returns and increased Negative Review % due to a packaging defect. As a result, the user can take the necessary steps to remediate these root causes to the sales decline.



Conversion Segments

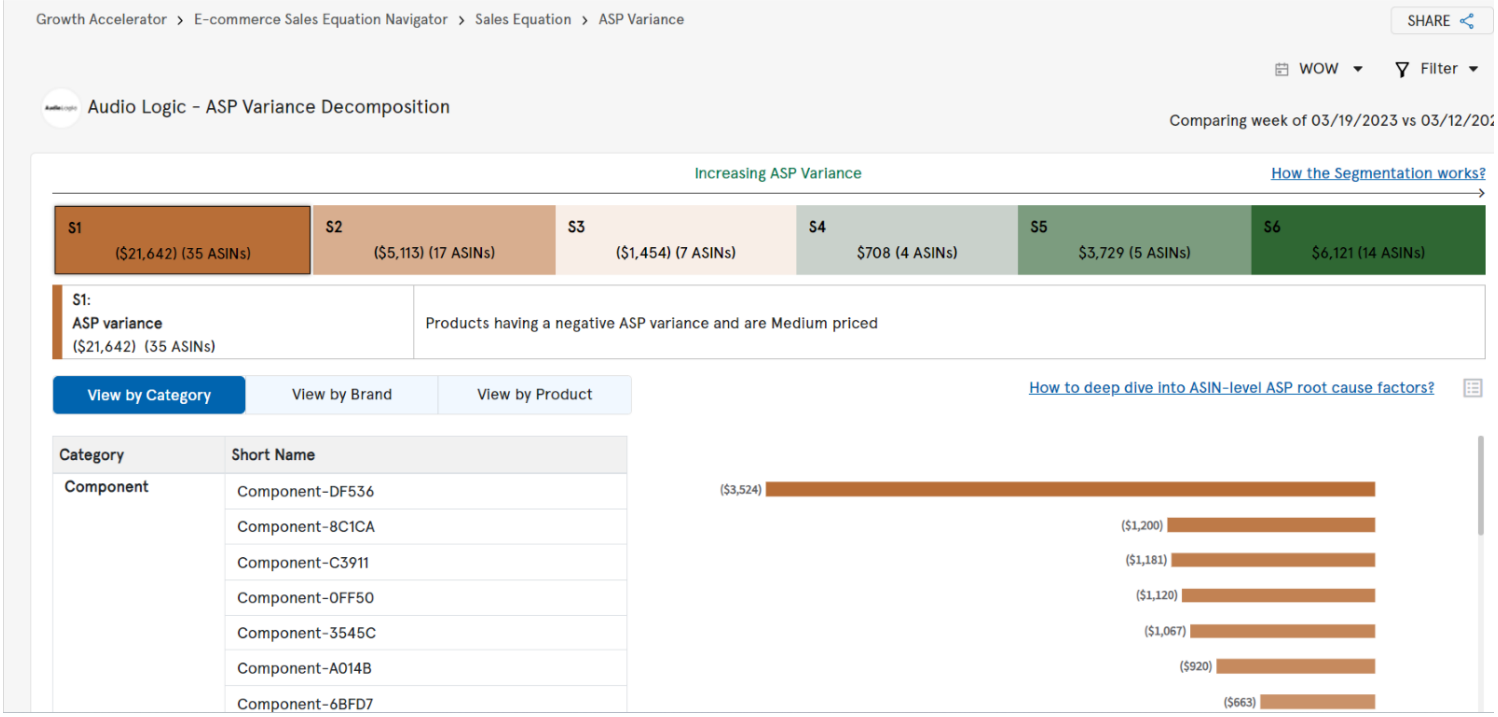
Similar to the Traffic Variance, Segments of the Conversion Variance are labeled under the names S1, S2, S3, etc. in order of the most negative variance to the most positive variance.



Click on any segment of the Conversion Decomposition to see product groups from poor to excellent sales performance.

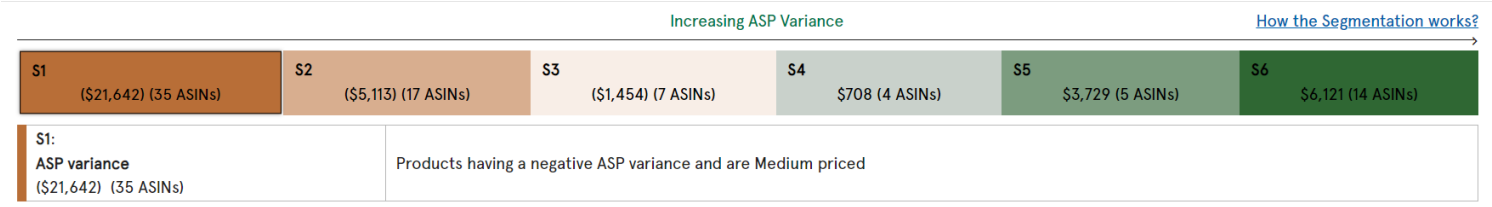
ASP Variance Page

On this page, products are placed into mutually exclusive Segments using a proprietary i2o Machine Learning Algorithm. Similar to Traffic and Conversion Variances, products in each ASP Segment share a similar price factor which has affected their sales in the selected time range (WoW, YoY, T4). The sum of variances across all segments is equal to the ASP Variance computed on the previous screen.



ASP Segments

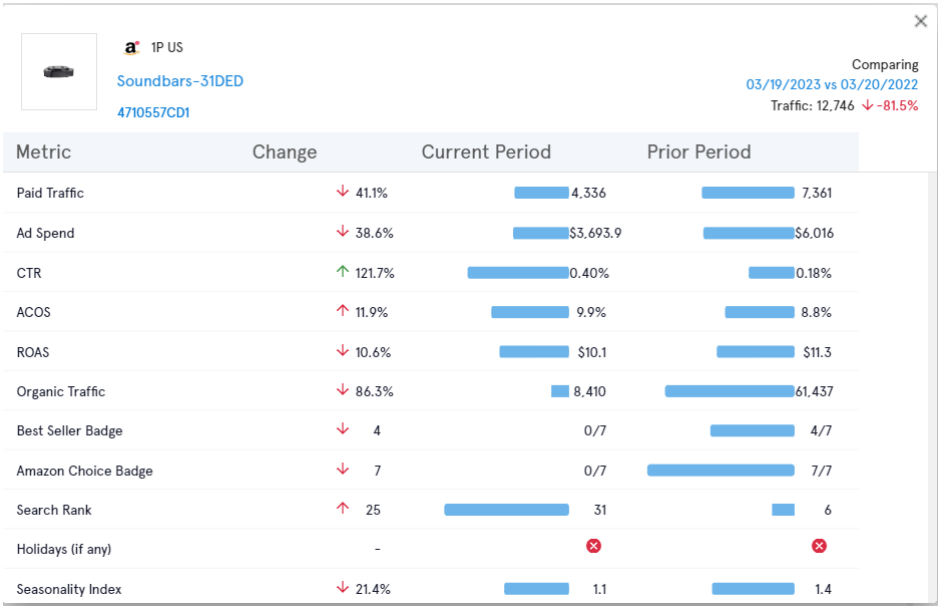
Similar to the Traffic and Conversion Variances, Segments of the ASP Variance are labeled under the names S1, S2, S3, etc. in order of the most negative variance to the most positive variance.



> Click on any segment of the ASP Decomposition to see product groups from poor to excellent sales performance.

ASIN Drilldown

The ASIN drilldown menu offers users a product-focused list of metrics impacting sales. There is a different menu for each part of the Sales Equation including Traffic, Conversion, and ASP.



Click on the ASIN bar and drill down further into the various KPIs affecting the respective selection. This is available for all Variance Pages including Traffic, Conversion, and ASP.

How to Use the E-commerce Sales Equation Navigator

To view the report:

1. Navigate to **Growth Accelerator > E-commerce Sales Equation Navigator**.
2. Click on a **Brand** or a product **Category** card to open the Sales Equation waterfall chart.
3. Select the drop-down next to WOW to choose from the below time periods:
    - Week Over Week
    - Year Over Year
    - T4 Avg.
4. Click on the inverted triangle next to **Filter**, to select and apply filters. You can apply only one filter at a time.
5. Click on any of the following three segments of the **Sales Equation** to see a product breakdown with attribution for change across the selected time periods.
    1. Traffic Variance
    2. Conversion Variance
    3. ASP Variance

How to Share/Export the E-commerce Sales Equation Report

The report on the **Sales Equation** page can be shared via email and can be exported as a PNG, JPEG or PDF file. Whereas the report on the Traffic, Conversion, or ASP variance page can be shared through an Email or downloaded as a PNG or a PDF file.

To share or export the report:

1. Navigate to **Growth Accelerator > E-commerce Sales Equation Navigator**.
2. Click on a brand or a product category card to open the **Sales Equation** waterfall chart.
3. Select the drop-down next to WOW to choose a time period for which you want to view the report.
4. Click on the inverted triangle next to **Filter**, to select and apply filters. You can apply only one filter at a time.
5. Click on the **Share** button to share the report over email. Multiple email IDs must be separated by comma.
6. Click on the ellipses to choose to export as PNG, JPEG, or PDF file format.

How to View the Traffic Variance Decomposition

To view Traffic Variance Decomposition report:

1. Navigate to **Growth Accelerator > E-commerce Sales Equation Navigator**.
2. Click on a brand or a product category card to open the **Sales Equation** waterfall chart.
3. Select the drop-down next to WOW to choose from the below time periods:
    - Week Over Week
    - Year Over Year
    - T4 Average
4. Click on the inverted triangle next to **Filter**, to select and apply filters. You can apply only one filter at a time.
5. Click on the **Traffic Variance** causal bar in the waterfall chart.
6. Click on a segment to view addition details. Segment S1 is the most negative whereas S5 is the most positive.
7. Filter the view by category, brand, or product.
8. Click on the variance bar which expands into all the underlying KPIs responsible for traffic and their change with respect to the compared period.

How to View the Conversion Variance Decomposition

To view Conversion Variance Decomposition report:

1. Navigate to **Growth Accelerator > E-commerce Sales Equation Navigator**.
2. Click on a brand or a product category card to open the **Sales Equation** waterfall chart.
3. Select the drop-down next to WOW to choose from the below time periods:
    - Week Over Week
    - Year Over Year
    - T4 Average
4. Click on the inverted triangle next to **Filter**, to select and apply filters. You can apply only one filter at a time.
5. Click on the **Conversion Variance** causal bar in the waterfall chart.
6. Click on a segment to view addition details Segment S1 is the most negative whereas S5 is the most positive.
7. Filter the view by category, brand, or product.

8. Click on the variance bar which expands into all the underlying KPIs responsible for traffic and their change with respect to the compared period.

How to View the ASP Variance Decomposition

To view ASP Variance Decomposition report:

1. Navigate to **Growth Accelerator > E-commerce Sales Equation Navigator**.
2. Click on a brand or a product category card to open the **Sales Equation** waterfall chart.
3. Select the drop-down next to WOW to choose from the below time periods:
    - Week Over Week
    - Year Over Year
    - T4 Average
4. Click on the inverted triangle next to **Filter**, to select and apply filters. You can apply only one filter at a time.
5. Click on the **ASP Variance** causal bar in the waterfall chart.
6. Click on a segment to view addition details Segment S1 is the most negative whereas S5 is the most positive.
7. Filter the view by category, brand, or product.
8. Click on the variance bar which expands into all the underlying KPIs responsible for traffic and their change with respect to the compared period.